Speech-Eval: finding speech delivery flaws, what, where and why

Technical documentation · Track C: Contrastive Speech Analytics & Temporal Flaw Grounding · Sriram MPS, IIT Mandi · October 2026

Code	Demo (precomputed)	Video	Dataset
github.com/mpssriram/speech-eval	HF Space: SriramMPS/Speech-Evalutaion	https://youtu.be/bbhAc6iyl7E	Google Drive (7.59 GB, SHA-256 pinned)

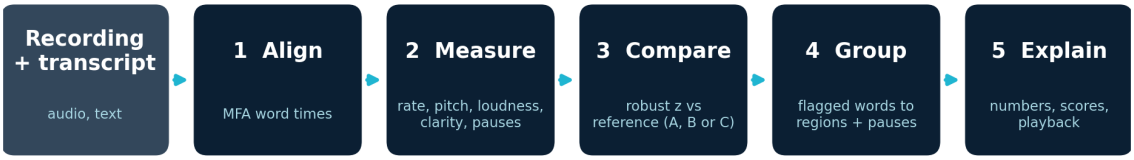
1. Summary

People who practise speaking usually get feedback such as “slow down” or “you sound flat”, which does not say *where* or *how much*. Speech-Eval compares a reading with a reference reading of the same text, word by word, and answers three questions for every flag: **what** went wrong, **where** (the exact words and seconds) and **why** (the measured value next to the reference value). The main detector, **v1**, was frozen before it saw the test set and stays the primary result.

- On 190 test recordings from unseen speakers, v1 found **40%** of injected flaws at the right place (95% CI 33–47%): awkward pauses 90%, rushing 17%. Precision is low (**12.5%**), and we say so.
- On the author's own takes it found all **4** planned pause flaws in Mode A and **none** of the 9 others, with about 10 false alarms per minute.
- We traced the pause false alarms to two causes and tested a pre-registered fix (a 52 ms timing floor). It removed 74–871 awkward-pause flags but found no new flaws, so v1 stays primary.
- The benchmark reproduces byte for byte from a fresh clone (563/563 outputs). The public demo shows these frozen results only.

> **How to read this report**
>
> Synthetic flaws are controlled edits, not human judgements. No independent listeners rated the recordings. Full derivations, detailed tables and every constant are in the separate **technical appendix**.

2. What we built



Every flag keeps its words, times, measured value, reference value and z-score.

Figure 1. The pipeline. Each step is deterministic and runs on CPU.

Mode	Compared with	Note
A	Expert readers of the same text	Strongest: spreads come from real readers
B	Your own earlier take	For rehearsal; one reference, so spreads are borrowed
C	Sentence norms from training speakers	No same-text reference; tips are not scored

Flaw	Measured by	Flag when
rushing / dragging	syllable rate in a 3-word window	much higher / much lower
monotone	pitch variation in semitones	much lower
too_quiet	loudness relative to your own median	much lower
mumbling	energy 2–6 kHz relative to 0–2 kHz	much lower
awkward_pause	gap before a word	much longer, and ≥ 0.45 s
missing_pause	gap before a word	much shorter; reference ≥ 0.3 s and yours ≤ 40% of it

The dashboard shows a score card (pace, pitch, pauses, volume, clarity), the waveform with every flagged region at its real width, plain-language explanations with the numbers, and **Yours** / **Baseline** buttons that play the same words from both recordings.

3. Dataset

No public dataset marks delivery flaws with exact times, so we built one. We took public-domain readings (Gettysburg Address, 15 LibriVox readers; O Captain! My Captain!, 13 readers; Kennedy at Rice; Reagan after Challenger) and inserted seven controlled effects at five strengths each at word-aligned positions: speed changes (×1.15 to ×2), pitch range reduced to 70–8%, gain of −3 to −18 dB, inserted pauses of 0.6–3 s, shortened pauses, and low-pass filtering.

Measurement	Value
Recordings	1,103 (30 originals + 1,073 edited), 36.3 hours; 13,812 files, each SHA-256 pinned
Split by speaker	18 train / 4 dev / 7 test; test voices never choose a threshold
Word aligner	MFA, median boundary error 52 ms (WhisperX 72 ms) on 37 visually marked words
Author's takes	8 consented recordings (CC BY 4.0), 13 planned flaws, positions frozen before detection

4. How it works: the key formulas

Each word window is measured, put on a log scale where a ratio matters (rate, pitch variation, pauses), and compared with the same word in the reference:

$$T_{\text{rate}}(x) = \ln x, \quad T_{\text{pitchvar}}(x) = \ln(x + 0.25), \quad T_{\text{pause}}(x) = \ln(x + 0.05), \quad T_{\text{loud}} = T_{\text{clarity}} = x$$

The reference centre μ is the median of the readers and their spread s is 1.4826 × MAD (the factor makes the MAD match a standard deviation for normal data). With one reader or a few readers, s can be zero, so the spread borrows from a pooled value counted as five extra readers:

$$\sigma^2 = \frac{n\, s^2 + n_0\, \sigma_{\text{pool}}^2}{n + n_0}, \qquad n_0 = 5$$

$$z = \frac{T(x) - \mu}{\sigma} \qquad (\text{rounded to 4 significant digits})$$

A word is flagged when |z| ≥ 3.5 in the flaw's direction (k chosen on development speakers only). Three or more flagged words in one sentence form a region; pauses are judged one boundary at a time and must also pass the length rules above. Each region gets a severity from 1 to 5 and costs points in proportion to how much speech it covers:

$$\text{penalty}_c = \sum_{r \in c} \text{sev}_r \cdot \frac{\text{end}_r - \text{start}_r}{D} \cdot 10$$

so severity-3 rushing over 10% of the speech gives a pace score of 7.0. Every formula is derived step by step in the technical appendix (Section A), including why each transform and constant has its value.

5. Results

Benchmark (frozen v1, test speakers, evaluated once)

Setting	Files	Precision	Recall [95% CI]	F1 [95% CI]	Clean alarms / min
A-LOSO (expert readers)	190	0.125	0.400 [0.328, 0.471]	0.190 [0.156, 0.223]	1.341
B-same-speaker (own original)	76	0.018	0.390 [0.289, 0.494]	0.035 [0.026, 0.045]	0.000

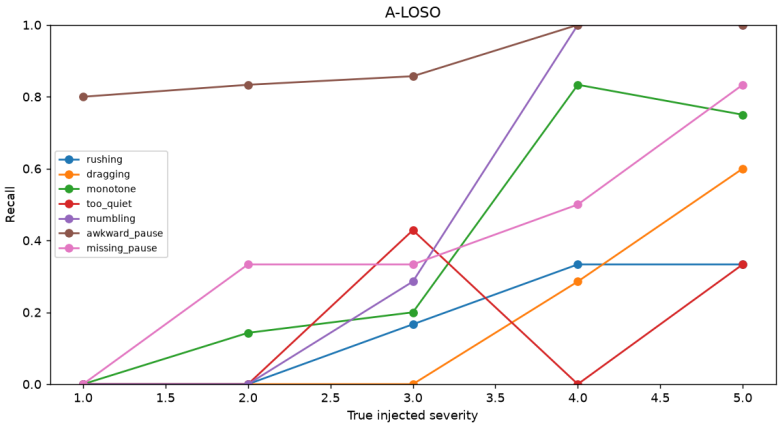


Figure 2. Recall against injected severity (A-LOSO). Awkward pauses are found even at severity 1.

Pauses are easiest because they are measured directly in seconds; rushing (recall 0.17) and too_quiet (0.18) are hardest. Mumbling has the best balance (F1 0.65, no clean alarms). Scores fall as flaws get worse (median Spearman −0.71 to −0.89). The development clean-alarm guard (≤ 0.5 per minute) was **not met** (1.78). A post-hoc v2 halved clean alarms but cut recall to 0.29, and Mode C (norms only) is much weaker than same-text comparison (recall 0.18 vs 0.32), so v1 stays primary. Per-type tables are in the technical appendix (Section B).

The author's own takes

Mode	Labels	Hits	Misses	False alarms	Per minute
A (expert readers)	13	4	9	77	9.6
B (own take 01)	13	3	10	60	11.2

All four hits are planned pause flaws: 1.57 s inside “larger | sense”, 1.56 s inside “were | daring”, and missing pauses after “here.” and “tragedy.”. None of the nine planned rushing, dragging, monotone, too-quiet or mumbling flaws produced a matching region.



Figure 3. Gettysburg take 02, Mode B. The selected awkward pause (29.90–31.47 s) matches the planned flaw exactly; the other 12 pause flags are false alarms.

6. Why mostly pause flags, and a fix

A read-only diagnosis (nothing retuned) found two separate causes:

- **Awkward pauses: a near-zero spread.** At most boundaries no reader pauses, so the pause spread is the numerical safeguard 0.001 and shrinks to about 0.0009. The real 1.57 s pause scores z = 3,810, and a 10 ms timing wobble scores z ≈ 20.
- **Missing pauses: a real difference.** 53 of the 77 Mode A false alarms are places where the author paused less than 40% as long as the expert readers. Whether listeners would call these flaws is unknown without ratings.
- **Other flaws: realistic spreads.** Planned rushing peaked at z = 5.3 but never formed a run of three flagged words. Every rule still fires on test probes; nothing is broken.

The fix, pre-registered. MFA's median boundary error is 52 ms, so pause differences smaller than that cannot be trusted. We set a floor on the pause spread equal to the log-scale effect of that error, committed the protocol (17a15e4) and code (ae38269) before any run, and ran each split once:

$$\sigma_{\text{floor}}(p) = \ln(p + 0.052 + 0.05) - \ln(p + 0.05) = \ln\left(1 + \frac{0.052}{p + 0.05}\right), \qquad \sigma_{\text{eff}} = \max(\sigma,\ \sigma_{\text{floor}})$$

Cohort	v1 F1	v1 + floor F1	Clean alarms / min	Flags removed
Test A-LOSO	0.190	0.205	1.341 → 1.118	74
Test B	0.035	0.061	0 → 0	871
Author's takes, A	0.085	0.094	8.65 → 7.89	9
Author's takes, B	0.079	0.103	n/a	18

Only awkward-pause flags were removed (missing-pause flags provably cannot be), three correct test matches were lost, and no non-pause flaw became a hit. Because the test set had been seen, v1 stays primary; the floor is a candidate for the next frozen version. Derivation and worked examples: the technical appendix (Sections C and D).

7. Public demo and reproducibility

The Hugging Face Space shows **precomputed, frozen v1 results** (19 views, 38 result files byte-identical to the report) and does not analyse uploads. This was deliberate:

- **Honest:** every number on the public page is a number in this report; a live server could drift from it.
- **Reliable:** no server to crash, time out or sleep; all 19 views are tested on the public URL.
- **Free:** live analysis needs MFA, Praat and a 16.8 GB Docker image; we found no free host that runs this reliably, and paid compute broke the project's zero-cost rule.
- **Private:** no one's voice is ever uploaded.
- **Why Hugging Face:** free static hosting with Git history and LFS for audio, anonymous access, and the place ML judges already look. The same files would work on any static host.

Live analysis runs unchanged on any machine with Docker (`git clone … && make up`). A fresh clone reproduced all **563 of 563** benchmark outputs byte for byte; the final check (download of the 7.59 GB dataset, verification, quick run) passed in 2,214 s, of which computation took 18.9 s. 344 tests pass.

8. Where this could help, and what comes next

Speech-Eval suits anyone who reads the same text more than once and wants to know exactly what changed. It would support a teacher or therapist, not replace one.

- **Students:** rehearse a talk or viva in Mode B and see which phrases changed between takes.
- **Language learners:** read aloud against model readers and fix pauses mid-phrase (delivery only, not accent).
- **People with Parkinson's disease:** reduced loudness, flat pitch and rate changes are typical [1, 2], and loudness-focused therapy has meta-analytic support [3]. Mode B could show where loudness or pitch variation dropped. **Not a medical device;** any such use needs a clinical study and a speech-language therapist.

If the project continues

Phase	Work	Done when
1. Validate	Independent listener ratings, human alignment check, fresh test speakers	Agreement measured
2. Better maths	Two-part pause model (how often readers pause, then how long); aligner noise in every spread; Benjamini–Hochberg false-alarm control per recording	Clean-alarm guard met on fresh data
3. Calibrate	Severity and scores fitted to listener ratings	Scores track what people hear
4. More languages	Per-language aligner and syllable counter; tone-aware pitch; own readers and thresholds (Hindi, Tamil, Telugu first)	Validation repeated per language
5. Product	Any text and spontaneous speech, progress over weeks, on-device privacy, teacher and therapist views	A class and clinic pilot

The formulas behind phase 2 are in the technical appendix (Section E).

9. Limitations

- Synthetic labels approximate flaws; they are not human judgements.
- Low precision (0.125 in Mode A, 0.018 in Mode B) and a failed development clean-alarm guard.
- No independent listener ratings; no human spot-check of alignment (reference boundaries were placed visually by Claude).
- Real takes: one speaker, six scored takes, 13 recorder-confirmed labels.
- v2 and the pause floor are post-hoc (test set already seen). English read speech only; rubric weights are design choices.
- Repository history keeps private workstation paths and metadata (no audio) of an excluded early test upload. The owner kept the history unchanged, because rewriting it would change the commit hashes cited as evidence that protocols came first.

10. Links, licences and credits

Item	Location
Code	https://github.com/mpssriram/speech-eval
Public demo (precomputed)	https://huggingface.co/spaces/SriramMPS/Speech-Evalutaion
Dataset v1 (7.59 GB)	Google Drive · SHA-256 a41a18482a72…e374a23b
Human takes (62.5 MB)	Google Drive · SHA-256 33e8e0e80a56…77658bd
Technical appendix	docs/report/technical-appendix.pdf
Demo video	https://youtu.be/bbhAc6iyl7E

Licences. Code MIT. Documentation, labels and the author's recordings CC BY 4.0. Reader and institutional audio US public domain. **Credits.** Author: Sriram MPS, IIT Mandi. Assistance: Claude (Anthropic) for planning, review, this report and the slides; Codex (OpenAI) for implementation. Tools: Montreal Forced Aligner, Praat/Parselmouth, librosa, CMU Pronouncing Dictionary, WhisperX, wav2vec 2.0. Thanks to the LibriVox volunteer readers.

- [1] Parkinson's disease-associated dysarthria: prevalence, impact and management strategies. doi:10.2147/JPRLS.S168090
- [2] Hypokinetic dysarthria in Parkinson's disease: a narrative review. PubMed 37899809
- [3] Yuan et al. (2020), LSVT for dysarthria in Parkinson's disease: systematic review and meta-analysis. doi:10.1111/ene.14399